

(Thus α is an F -isomorphism in the sense of §4.)

PROOF. If $u \in \ker \alpha$, then 6.2 with $l = \infty$ and $k = 1$ shows that $u = pv$ for some $v \in H(\infty)$. Hence $u^k = p^k v^k = 0$ for large k , since $H(\infty)$ is annihilated by some p^k . This proves (i), and (ii) follows from 6.4 and 6.5. $\square$

PROOF OF 6.1. The map $\alpha: H(\infty) \rightarrow H(1)$ is a ring homomorphism (taking 1 to 1). This follows from the fact that the map $\hat{H}^*(\Gamma, \mathbb{Z}) \rightarrow \hat{H}^*(\Gamma, \mathbb{Z}_p)$ is a ring homomorphism which maps all primary components to zero except $\hat{H}^*(\Gamma, \mathbb{Z})_{(p)}$. Thus α takes invertible elements to invertible elements, whence the “only if” part of the proposition. Conversely, suppose $H(1)$ contains an invertible element u of non-zero degree. Raising u to a power if necessary, we can assume (6.6(ii)) that $u, u^{-1} \in \text{im } \alpha$. Let $u = \alpha(\tilde{u})$ and $u^{-1} = \alpha(\tilde{v})$. Then $\alpha(\tilde{u}\tilde{v}) = 1$, so 6.6(i) implies that $\tilde{u}\tilde{v} = 1 - x$ with x nilpotent. But then $1 - x$ is invertible, with inverse $1 + x + x^2 + \dots$, so $\tilde{u}$ is invertible and Γ has p -periodic cohomology. $\square$

We now give the main result of this section:

(6.7) Theorem. *The following conditions are equivalent:*

- (i) Γ has p -periodic cohomology.
- (ii) There exist integers i and d with $d \neq 0$ such that $\hat{H}^i(\Gamma, M)_{(p)} \approx \hat{H}^{i+d}(\Gamma, M)_{(p)}$ for all Γ -modules M .
- (iii) Every finite subgroup of Γ has p -periodic cohomology.
- (iv) Every elementary abelian p -subgroup of Γ has rank ≤ 1 .

PROOF. (i) $\Rightarrow$ (ii) trivially. If (ii) holds, then (ii) holds for every finite subgroup of Γ by Shapiro’s lemma. But this implies (iii) by the analogue of VI.9.1 for p -periodicity. Since (iii) $\Leftrightarrow$ (iv) by VI.9.7, it remains to prove (iii) $\Rightarrow$ (i).

Let $\mathfrak{F}$ and $\mathfrak{H}^*(\Gamma)$ be as in 4.4, with $\mathbb{Z}_p$ as coefficient module. If (iii) holds, then I claim that we can find a family $u = (u_H)_{H \in \mathfrak{F}}$ of invertible elements $u_H \in \hat{H}^d(H) = \hat{H}^d(H, \mathbb{Z}_p)$ such that $u \in \mathfrak{H}^d(\Gamma)$. To see this, note first that the elements of $\mathfrak{F}$ fall into finitely many isomorphism classes; for if $\Gamma' \subseteq \Gamma$ is a torsion-free normal subgroup of finite index, then every $H \in \mathfrak{F}$ is isomorphic to a subgroup of the finite group Γ/Γ' . Now choose for each $H \in \mathfrak{F}$ an invertible element $v_H \in \hat{H}^d(H)$ of degree $d_H \neq 0$. By what we have just observed, we can certainly assume that only finitely many distinct degrees d_H occur. Raising each v_H to a suitable power, if necessary, we can then arrange that all the d_H are equal to a single integer d .

Now let $c: H_1 \hookrightarrow H_2$ be an embedding of finite subgroups given by conjugation by some element of Γ . Then v_{H_1} and $c^*v_{H_2}$ are two generators of $\hat{H}^d(H_1) \approx \hat{H}^d(H_1)$, hence $c^*v_{H_2} = \lambda v_{H_1}$ for some non-zero scalar $\lambda \in \mathbb{Z}_p$. Since $\lambda^{p-1} = 1$, it follows that $v_{H_1}^{p-1} = (c^*v_{H_2})^{p-1} = c^*(v_{H_2}^{p-1})$. Thus if we set $u_H = v_H^{p-1}$, we have $(u_H) \in \mathfrak{H}^*(\Gamma)$. Clearly (u_H^{-1}) is also in $\mathfrak{H}^*(\Gamma)$, so $\mathfrak{H}^*(\Gamma)$ contains an invertible element of non-zero degree. In view of the F -isomorphism $\hat{H}^*(\Gamma) \rightarrow \mathfrak{H}^*(\Gamma)$ of 4.6, it follows exactly as in the proof of 6.1 that $\hat{H}^*(\Gamma) = \hat{H}^*(\Gamma, \mathbb{Z}_p)$ contains an invertible element of non-zero degree. $\square$

Remark. A weaker version of the implication (iii) $\Rightarrow$ (i) was proved by Venkov [1965], but not in the language of Farrell cohomology—he spoke instead of the ordinary cohomology being periodic in sufficiently high dimensions. Restated in terms of Farrell cohomology, his result said: if $\hat{H}^d(\Gamma)$ contains an element u whose restriction to $\hat{H}^*(H)$ is invertible for every $H \in \mathfrak{F}$, then u is invertible. (His proof, like ours, was based on the multiplicative structure in the spectral sequence 4.1.) The missing ingredient which was needed to go from Venkov's result to Theorem 6.7 was Quillen's observation 4.6.

EXERCISE

If 6.7(ii) holds and the isomorphism is natural, show that $\hat{H}^*(\Gamma, \mathbb{Z})_{(p)}$ contains an invertible element of degree d . [Hint: By dimension-shifting, 6.7(ii) holds for all i , with isomorphisms that are compatible with connecting homomorphisms. Show that the composite $H^i \rightarrow \hat{H}_{(p)}^i \rightarrow \hat{H}_{(p)}^{i+d}$ is necessarily given (up to sign) by cup product with the image of $1 \in H^0$.]

7 $\hat{H}^*(\Gamma)$ and the Ordered Set of Finite Subgroups of Γ

In this section we will prove analogues for Farrell cohomology of Theorems IX.11.1 and IX.13.1.

If Γ operates on a partially ordered set S such that $|S|$ is finite dimensional, then we set

$$\hat{H}_\Gamma^*(S) = \hat{H}_\Gamma^*(|S|).$$

As usual, it is understood here that there is an arbitrary Γ -module of coefficients. Recall that for any finite dimensional Γ -complex X there is a canonical map

$$\hat{H}^*(\Gamma) = \hat{H}_\Gamma^*(\text{pt.}) \rightarrow \hat{H}_\Gamma^*(X),$$

induced by the map $X \rightarrow \text{pt.}$

(7.1) Theorem. *Let Γ be a group such that $\text{vcd } \Gamma < \infty$, and let S be the set of non-trivial finite subgroups of Γ . Then the canonical map*

$$\hat{H}^*(\Gamma) \rightarrow \hat{H}_\Gamma^*(S)$$

is an isomorphism.

PROOF. As in the proof of IX.11.1, let X be a finite dimensional, contractible, admissible, proper Γ -complex with contractible fixed-point sets X^H ($H \in S$), and let $X_0 = \bigcup_{H \in S} X^H$. Then we have

$$\begin{aligned} \hat{H}^*(\Gamma) &\approx \hat{H}_\Gamma^*(X) && \text{because } X \text{ is contractible} \\ &\approx \hat{H}_\Gamma^*(X_0) && \text{because } \Gamma \text{ acts freely in } X - X_0 \text{ (cf. VII.10.1)} \\ &\approx \hat{H}_\Gamma^*(S) && \text{by IX.11.2.} \end{aligned}$$

To see that the composite isomorphism is given by the canonical map $\hat{H}^*(\Gamma) \rightarrow \hat{H}_\Gamma^*(S)$, consider the diagram.

$$(*) \quad \begin{array}{ccc} & \hat{H}^*(\Gamma) & \\ \swarrow \approx & \downarrow & \searrow \\ \hat{H}_\Gamma^*(X) & \approx \hat{H}_\Gamma^*(X_0) & \approx \hat{H}_\Gamma^*(S), \end{array}$$

where all maps coming from $\hat{H}^*(\Gamma)$ are the canonical ones. The left-hand triangle obviously commutes. For the right-hand triangle, we must go back to IX.11.2 and note that, in the notation of the latter, the square

$$\begin{array}{ccc} C & \longrightarrow & C(S) \\ \downarrow & & \downarrow \\ C(Z) & \longrightarrow & C(\text{pt.}) \end{array}$$

commutes. This is easily verified. Thus $(*)$ commutes, whence the theorem. $\square$

As in Chapter IX, there is a p -local version of 7.1 which is more useful. Its proof will be based on the following analogue of IX.11.5:

(7.2) Proposition. *Let Z be a finite dimensional admissible Γ -complex and let p be a prime such that Z^H is acyclic for every non-trivial finite p -subgroup of Γ . Then the canonical map*

$$\hat{H}^*(\Gamma)_{(p)} \rightarrow \hat{H}_\Gamma^*(Z)_{(p)}$$

is an isomorphism.

PROOF. Assume first that every finite subgroup of Γ is a p -group. We can then apply the method of proof of 7.1 to Γ acting on $X \times Z$ (with X as above), and we obtain an isomorphism $\hat{H}_\Gamma^*(X \times Z) \approx \hat{H}_\Gamma^*(S)$ such that the triangle

$$\begin{array}{ccc} & \hat{H}^*(\Gamma) & \\ \swarrow & & \searrow \approx \\ \hat{H}_\Gamma^*(X \times Z) & \approx & \hat{H}_\Gamma^*(S) \end{array}$$

commutes. On the other hand, the projection $X \times Z \rightarrow Z$ yields

$$\begin{array}{ccc} & \hat{H}^*(\Gamma) & \\ & \swarrow \quad \searrow & \\ \hat{H}^*_\Gamma(Z) & \xrightarrow{\approx} & \hat{H}^*_\Gamma(X \times Z). \end{array}$$

We therefore have

$$\hat{H}^*(\Gamma) \xrightarrow{\approx} \hat{H}^*_\Gamma(Z)$$

in this case.

In the general case, choose a subgroup $\Gamma' \subseteq \Gamma$ of finite index such that $(\Gamma: \Gamma')$ is relatively prime to p and every finite subgroup of Γ' is a p -group (cf. proof of IX.9.2). Using restriction and transfer maps in the usual way (cf. §III.10), we have a natural embedding of $\hat{H}^*_\Gamma(W)_{(p)}$ (for any W) as a direct summand of $\hat{H}^*_{\Gamma'}(W)$. In particular, taking $W = Z$ and $W = \text{pt.}$, we see that the canonical map

$$\hat{H}^*(\Gamma)_{(p)} \rightarrow \hat{H}^*_\Gamma(Z)_{(p)}$$

is a direct summand of the canonical map

$$\hat{H}^*(\Gamma') \rightarrow \hat{H}^*_{\Gamma'}(Z).$$

The latter being an isomorphism by the previous paragraph, it follows that the former is also an isomorphism. $\square$

We can now prove the main result of this section:

(7.3) Theorem. *Let Γ be a group such that $\text{vcd } \Gamma < \infty$, let p be a prime, and let $\mathcal{A}$ be the ordered set of non-trivial elementary abelian p -subgroups of Γ . Then*

$$\hat{H}^*(\Gamma)_{(p)} \xrightarrow{\approx} \hat{H}^*_\Gamma(\mathcal{A})_{(p)}.$$

PROOF. Apply 7.2 with $Z = |\mathcal{A}|$ (cf. proof of IX.13.1). $\square$

One special case where $\hat{H}^*_\Gamma(\mathcal{A})$ can be easily understood is that where $\mathcal{A}$ is discrete. In that case 7.3 yields:

(7.4) Corollary. *Suppose that every elementary abelian p -subgroup of Γ has rank ≤ 1 . Then*

$$\hat{H}^*(\Gamma)_{(p)} \xrightarrow{\approx} \prod_{P \in \mathcal{P}} \hat{H}^*(N(P))_{(p)},$$

where $\mathcal{P}$ is a set of representatives for the conjugacy classes of subgroups of Γ of order p .

For a concrete example where 7.4 leads to explicit cohomology calculations, see Brown [1976]. (These calculations, however, which involve $\Gamma = SL_3(\mathbb{Z})$, have been superseded by the calculations of Soulé [1978].) For an example of 7.3 with $\dim \mathcal{A} > 0$, see Brown [1979]. This time $\Gamma = SL_3(\mathbb{Z}[1/2])$, $p = 2$, and $\dim \mathcal{A} = 1$.

Remark. Since $\hat{H}^i(\Gamma) = H^i(\Gamma)$ for $i > \text{vcd } \Gamma$, the canonical map

$$H^i(\Gamma)_{(p)} \rightarrow \prod_{P \in \mathcal{P}} H^i(N(P))_{(p)}$$

is an isomorphism for $i > \text{vcd } \Gamma$ under the hypotheses of 7.4. It is natural to ask, by analogy with similar situations in algebraic topology, whether the map is surjective for $i = \text{vcd } \Gamma$. It can be shown that this question, whose answer is not known, is closely related to the question asked in §VIII.11 as to whether one can always find a proper, contractible Γ -complex of dimension equal to $\text{vcd } \Gamma$.

EXERCISE

Show that the components of the map in 7.4 are restriction maps

$$\hat{H}^*(\Gamma)_{(p)} \rightarrow \hat{H}^*(N(P))_{(p)}.$$

References

- G. Almkvist [1973], Endomorphisms of finitely generated projective modules over a commutative ring, *Ark. Mat.* **11** (1973), 263–301.
- R. C. Alperin and P. B. Shalen [1982], Linear groups of finite cohomological dimension, *Invent. Math.* **66** (1982), 89–98.
- A. Ash [1977], Deformation retracts with lowest possible dimension of arithmetic quotients of self-adjoint homogeneous cones, *Math. Ann.* **225** (1977), 69–76.
- M. F. Atiyah and I. G. Macdonald [1969], *Introduction to commutative algebra*, Addison-Wesley, Reading, Mass., 1969.
- H. Bass [1976], Euler characteristics and characters of discrete groups, *Invent. Math.* **35** (1976), 155–196.
- H. Bass [1979], Traces and Euler characteristics, Homological group theory (C. T. C. Wall, ed.), *London Math. Soc. Lecture Notes* **36**, Cambridge University Press, Cambridge, 1979, 105–136.
- R. Bieri [1976], Homological dimension of discrete groups, Queen Mary College Mathematics Notes, London, 1976.
- R. Bieri and B. Eckmann [1973], Groups with homological duality generalizing Poincaré duality, *Invent. Math.* **20** (1973), 103–124.
- R. Bieri and B. Eckmann [1974], Finiteness properties of duality groups, *Comment. Math. Helv.* **49** (1974), 74–83.
- A. Borel et al. [1960], Seminar on transformation groups, *Ann. of Math. Studies* **46**, Princeton, 1960.
- A. Borel and J.-P. Serre [1974], Corners and arithmetic groups, *Comment. Math. Helv.* **48** (1974), 244–297.
- A. Borel and J.-P. Serre [1976], Cohomologie d'immeubles et de groupes S-arithmétiques, *Topology* **15** (1976), 211–232.
- R. Brauer [1926], Über Zusammenhänge zwischen arithmetischen und invariantentheoretischen Eigenschaften von Gruppen linearer Substitutionen, *Sitzungsber. Preuss. Akad. Wiss.* (1926), 410–416. Collected papers, vol. 1, 5–11.
- G. E. Bredon [1972], *Introduction to compact transformation groups*, Academic Press, New York, 1972.
- K. S. Brown [1974], Euler characteristics of discrete groups and G-spaces, *Invent. Math.* **27** (1974), 229–264.

- K. S. Brown [1975a], Homological criteria for finiteness, *Comment. Math. Helv.* **50** (1975), 129–135.
- K. S. Brown [1975b], Euler characteristics of groups: The p -fractional part, *Invent. Math.* **29** (1975), 1–5.
- K. S. Brown [1976], High dimensional cohomology of discrete groups, *Proc. Nat. Acad. Sci. U.S.A.* **73** (1976), 1795–1797.
- K. S. Brown [1979], Groups of virtually finite dimension, Homological group theory (C. T. C. Wall, ed.), *London Math. Soc. Lecture Notes* **36**, Cambridge University Press, Cambridge, 1979, 27–70.
- K. S. Brown [1982], Complete Euler characteristics and fixed-point theory, *J. Pure Appl. Algebra* **24** (1982), 103–121.
- K. S. Brown and P. J. Kahn [1977], Homotopy dimension and simple cohomological dimension of spaces, *Comment. Math. Helv.* **52** (1977), 111–127.
- W. Burnside [1911], *Theory of groups of finite order (2nd edition)*, Cambridge University Press, Cambridge, 1911. (Reprinted by Dover, New York, 1955.)
- H. Cartan et al. [1954/55], *Algèbres d'Eilenberg-MacLane et homotopie*, Séminaire Henri Cartan 1954/55, École Normale Supérieure, Paris.
- H. Cartan and S. Eilenberg [1956], *Homological algebra*, Princeton University Press, Princeton, 1956.
- I. M. Chiswell [1976a], Exact sequences associated with a graph of groups, *J. Pure Appl. Algebra* **8** (1976), 63–74.
- I. M. Chiswell [1976b], Euler characteristics of groups, *Math. Z.* **147** (1976), 1–11.
- D. E. Cohen [1972], Groups with free subgroups of finite index, Conference on group theory, *Lecture Notes in Math.* **319**, Springer-Verlag, Berlin-Heidelberg-New York, 1972, 26–44.
- D. E. Cohen [1978], *Combinatorial group theory: A topological approach*, Queen Mary College Mathematics Notes, London, 1978.
- J. Cohen [1972], Poincaré 2-complexes—I, *Topology* **11** (1972), 417–419.
- H. S. M. Coxeter and W. O. J. Moser [1980], *Generators and relations for discrete groups*, 4th ed., Springer-Verlag, Berlin-Heidelberg-New York, 1980.
- A. Dold [1972], *Lectures on algebraic topology*, Springer-Verlag, Berlin-Heidelberg-New York, 1972.
- M. J. Dunwoody [1972], Relation modules, *Bull. London Math. Soc.* **4** (1972), 151–155.
- M. J. Dunwoody [1979], Accessibility and groups of cohomological dimension one, *Proc. London Math. Soc.* (3) **38** (1979), 193–215.
- E. Dyer and A. T. Vasquez [1973], Some small aspherical spaces, *J. Austral. Math. Soc.* **16** (1973), 332–352.
- B. Eckmann [1953], Cohomology of groups and transfer, *Ann. of Math.* **58** (1953), 481–493.
- B. Eckmann and H. Müller [1980], Poincaré duality groups of dimension two, *Comment. Math. Helv.* **55** (1980), 510–520.
- S. Eilenberg [1947], Homology of spaces with operators, I, *Trans. Amer. Math. Soc.* **61** (1947), 378–417; errata, **62** (1947), 548.
- S. Eilenberg and T. Ganea [1957], On the Lusternik-Schnirelmann category of abstract groups, *Ann. of Math.* **65** (1957), 517–518.
- S. Eilenberg and S. MacLane [1945], Relations between homology and homotopy groups of spaces, *Ann. of Math.* (2) **46** (1945), 480–509.
- S. Eilenberg and S. MacLane [1947], Cohomology theory in abstract groups, II, Group extensions with a non-abelian kernel, *Ann. of Math* (2) **48** (1947), 326–341.
- S. Eilenberg and N. Steenrod [1952], *Foundations of algebraic topology*, Princeton University Press, Princeton, 1952.
- F. T. Farrell [1975], Poincaré duality and groups of type (FP), *Comment. Math. Helv.* **50** (1975), 187–195.
- F. T. Farrell [1977], An extension of Tate cohomology to a class of infinite groups, *J. Pure Appl. Algebra* **10** (1977), 153–161.

- J. Fischer, A. Karrass, and D. Solitar [1972], On one-relator groups having elements of finite order, *Proc. Amer. Math. Soc.* **33** (1972), 297–301.
- R. Godement [1958], *Topologie algébrique et théorie des faisceaux*, Hermann, Paris, 1958.
- O. Goldman [1961], Determinants in projective modules, *Nagoya Math. J.* **18** (1961), 27–36.
- D. H. Gottlieb [1965], A certain subgroup of the fundamental group, *Amer. J. Math.* **87** (1965), 840–856.
- A. Grothendieck [1957], Sur quelques points d'algèbre homologique, *Tôhoku Math. J.* (2) **9** (1957), 119–221.
- K. W. Gruenberg [1970], Cohomological topics in group theory, *Lecture Notes in Math.* **143**, Springer-Verlag, Berlin-Heidelberg-New York, 1970.
- M. Hall, Jr. [1959], *The theory of groups*, MacMillan, New York, 1959.
- G. Harder [1971], A Gauss-Bonnet formula for discrete arithmetically defined groups, *Ann. Sci. École Norm. Sup* (4) **4** (1971), 409–455.
- A. Hattori [1965], Rank element of a projective module, *Nagoya J. Math.* **25** (1965), 113–120.
- G. Hochschild [1965], *The structure of Lie groups*, Holden-Day, San Francisco, 1965.
- G. Hochschild and J-P. Serre [1953], Cohomology of group extensions, *Trans. Amer. Math. Soc.* **74** (1953), 110–134.
- H. Hopf [1942], Fundamentalgruppe und zweite Bettische Gruppe, *Comment. Math. Helv.* **14** (1942), 257–309.
- J. F. P. Hudson [1969], *Piecewise linear topology*, Benjamin, New York, 1969.
- N. J. S. Hughes [1951], The use of bilinear mappings in the classification of groups of class 2, *Proc. Amer. Math. Soc.* **2** (1951), 742–747.
- W. Hurewicz [1936], Beiträge zur Topologie der Deformationen. IV. Asphärische Räume, *Nederl. Akad. Wetensch. Proc.* **39** (1936), 215–224.
- S. Illman [1978], Smooth equivariant triangulations of G -manifolds for G a finite group, *Math. Ann.* **233** (1978), 199–220.
- S. Jackowski [1978], The Euler class and periodicity of group cohomology, *Comment. Math. Helv.* **53** (1978), 643–650.
- A. Karrass, A. Pietrowski, and D. Solitar [1973], Finitely generated groups with a free subgroup of finite index, *J. Austral. Math. Soc.* **16** (1973), 458–466.
- R. C. Kirby and L. C. Siebenmann [1969], On the triangulation of manifolds and the Hauptvermutung, *Bull. Amer. Math. Soc.* **75** (1969), 742–749.
- J. Lehner [1964], *Discontinuous groups and automorphic functions*, Mathematical surveys no. 8, Amer. Math. Soc., Providence, 1964.
- A. T. Lundell and S. Weingram [1969], *The topology of CW complexes*, Van Nostrand Reinhold, New York, 1969.
- R. C. Lyndon [1950], Cohomology theory of groups with a single defining relation, *Ann. of Math.* **52** (1950), 650–665.
- R. C. Lyndon and P. E. Schupp [1977], *Combinatorial group theory*, Springer-Verlag, Berlin-Heidelberg-New York, 1977.
- S. MacLane [1949], Cohomology theory in abstract groups, III, Operator homomorphisms of kernels, *Ann. of Math.* (2) **50** (1949), 736–761.
- S. MacLane [1963], *Homology*, Springer-Verlag, Berlin, 1963.
- S. MacLane [1978], Origins of the cohomology of groups, *Enseign. Math.* (2) **24** (1978), 1–29.
- S. MacLane [1979], Historical note, *J. Algebra* **60** (1979), 319–320.
- I. Madsen, C. B. Thomas, and C. T. C. Wall [1976], The topological space form problem—II: Existence of free actions, *Topology* **15** (1976), 375–382.
- A. I. Malcev [1949], On a class of homogeneous spaces, *Izv. Akad. Nauk SSSR Ser. Mat.* **13** (1949), 9–32. (English translation: *Amer. Math. Soc. Transl. No.* **39** (1951).)

- W. S. Massey [1967], *Algebraic topology: an introduction*, Harcourt, Brace & World, New York, 1967. (Reprinted by Springer-Verlag.)
- C. Miller [1952], The second homology group of a group; relations among commutators, *Proc. Amer. Math. Soc.* **3** (1952), 588–595.
- J. Milnor [1957a], Groups which act on S^n without fixed points, *Amer. J. Math.* **79** (1957), 623–630.
- J. Milnor [1957b], The geometric realization of a semi-simplicial complex, *Ann. of Math.* (2) **65** (1957), 357–362.
- J. Milnor [1963], Morse theory, *Ann. of Math. Studies* **51**, Princeton University Press, Princeton, 1963.
- J. Milnor [1971], Introduction to algebraic K-theory, *Ann. of Math. Studies* **72**, Princeton University Press, Princeton, 1971.
- M. S. Montgomery [1969], Left and right inverses in group algebras, *Bull. Amer. Math. Soc.* **75** (1969), 539–540.
- J. Munkres [1966], Elementary differential topology, revised ed., *Ann. of Math. Studies* **54**, Princeton University Press, Princeton, 1966.
- C. D. Papakyriakopoulos [1957], On Dehn's lemma and the asphericity of knots, *Ann. of Math.* **66** (1957), 1–26.
- D. S. Passman [1977], *The algebraic structure of group rings*, Wiley, New York, 1977.
- H. Poincaré [1904], Cinquième complément à l'Analysis situs, *Rend. Circ. Mat. Palermo* **18** (1904), 45–110. (*Oeuvres*, Gauthier-Villars (1953), t. VI, pp. 435–498.)
- D. Quillen [1971], The spectrum of an equivariant cohomology ring, I, II, *Ann. of Math.* **94** (1971), 549–572 and 573–602.
- D. Quillen [1973], Finite generation of the groups K_i of rings of algebraic integers, Algebraic K-theory I, *Lecture Notes in Math.* **341**, Springer-Verlag, Berlin-Heidelberg-New York, 1973, 179–198.
- D. Quillen [1978], Homotopy properties of the poset of non-trivial p -subgroups of a group, *Advances in Math.* **28** (1978), 101–128.
- H. R. Schneebeli [1978], On virtual properties of group extensions, *Math. Z.* **159** (1978), 159–167.
- O. Schreier [1926], Über die Erweiterungen von Gruppen, I, *Monatsh. Math. u. Phys.* **34** (1926), 165–180.
- H. Schubert [1968], *Topology*, Allyn and Bacon, Boston, 1968.
- I. Schur [1902], Neuer Beweis eines Satzes über endliche Gruppen, *Sitzungsber. Preuss. Akad. Wiss.* (1902), 1013–1019. *Gesammelte Abhandlungen*, I, 79–85.
- I. Schur [1904], Über die Darstellung der endlichen Gruppen durch gebrochene lineare Substitutionen, *J. Reine Angew. Math.* **127** (1904), 20–50. *Gesammelte Abhandlungen*, I, 86–116.
- G. P. Scott [1974], An embedding theorem for groups with a free subgroup of finite index, *Bull. London Math. Soc.* **6** (1974), 304–306.
- G. P. Scott and C. T. C. Wall [1979], Topological methods in group theory, Homological group theory (C. T. C. Wall, ed.), *London Math. Soc. Lecture Notes* **36**, Cambridge University Press, Cambridge, 1979, 137–203.
- J-P. Serre [1965], Algèbre locale; multiplicités (2nd edition), *Lecture Notes in Math.* **11**, Springer-Verlag, Berlin-Heidelberg-New York, 1965.
- J-P. Serre [1968], *Corps locaux*, Hermann, Paris, 1968.
- J-P. Serre [1971], Cohomologie des groupes discrets, *Ann. of Math. Studies* **70** (1971), 77–169.
- J-P. Serre [1977a], Arbres, amalgames, SL_2 , *Astérisque* **46** (1977).
- J-P. Serre [1977b], *Linear representations of finite groups*, Springer-Verlag, New York, 1977.
- J-P. Serre [1979], Arithmetic groups, Homological group theory (C. T. C. Wall, ed.), *London Math. Soc. Lecture Notes* **36**, Cambridge University Press, Cambridge, 1979, 105–136.

- C. L. Siegel [1971], *Topics in complex function theory*, vol. II, Wiley-Interscience, New York, 1971.
- C. Soulé [1978], The cohomology of $SL_3(\mathbb{Z})$, *Topology* **17** (1978), 1–22.
- E. H. Spanier [1966], *Algebraic Topology*, McGraw-Hill, New York, 1966.
- J. R. Stallings [1963], A finitely presented group whose 3-dimensional integral homology is not finitely generated, *Amer. J. Math.* **85** (1963), 541–543.
- J. R. Stallings [1965a], Homology and central series of groups, *J. Algebra* **2** (1965), 170–181.
- J. R. Stallings [1965b], Centerless groups—an algebraic formulation of Gottlieb’s theorem, *Topology* **4** (1965), 129–134.
- J. R. Stallings [1968], On torsion-free groups with infinitely many ends, *Ann. of Math.* **88** (1968), 312–334.
- J. R. Stallings [unpublished], An extension theorem for Euler characteristics of groups, unpublished.
- U. Stambach [1966], Anwendungen der Homologietheorie der Gruppen auf Zentralreihen und auf Invarianten von Präsentierungen, *Math. Z.* **94** (1966), 157–177.
- U. Stambach [1973], Homology in group theory, *Lecture Notes in Math.* **359**, Springer-Verlag, Berlin-Heidelberg-New York, 1973.
- R. Strebel [1976], A homological finiteness criterion, *Math. Z.* **151** (1976), 263–275.
- R. Strebel [1977], A remark on subgroups of infinite index in Poincaré duality groups, *Comment. Math. Helv.* **52** (1977), 317–324.
- U. Stuhler [1980], Homological properties of certain arithmetic groups in the function field case, *Invent. Math.* **57** (1980), 263–281.
- M. Suzuki [1955], On finite groups with cyclic Sylow subgroups for all odd primes, *Amer. J. Math.* **77** (1955), 657–691.
- R. G. Swan [1960a], A new method in fixed point theory, *Comment. Math. Helv.* **34** (1960), 1–16.
- R. G. Swan [1960b], Induced representations and projective modules, *Ann. of Math.* **71** (1960), 552–578.
- R. G. Swan [1969], Groups of cohomological dimension one, *J. Algebra* **12** (1969), 585–601.
- O. Teichmüller [1940], Über die sogenannte nichtcommutative Galoische Theorie und die Relation $\xi_{\lambda, \mu, \nu} \xi_{\lambda, \mu, \nu} \pi \xi_{\mu, \nu, \pi} = \xi_{\lambda, \mu, \nu \pi} \xi_{\lambda \mu, \nu, \pi}$, *Deutsche Math.* **5** (1940) 138–149.
- B. B. Venkov [1965], On homologies of groups of units in division algebras, *Trudy Mat. Inst. Steklov* **80** (1965), 66–89. [English translation: *Proc. Steklov Inst. Math.* **80** (1965), 73–100.]
- G. Voronoi [1907], Nouvelles applications des paramètres continus à la théorie des formes quadratiques, I, *J. Reine Angew. Math.* **133** (1907), 97–178.
- C. T. C. Wall [1961], Rational Euler characteristics, *Proc. Cambridge Philos. Soc.* **57** (1961), 182–183.
- C. T. C. Wall [1965], Finiteness conditions for CW-complexes, *Ann. of Math.* **81** (1965), 56–69.
- C. T. C. Wall [1966], Finiteness conditions for CW-complexes II, *Proc. Royal Soc. A* **295** (1966), 129–139.
- C. T. C. Wall (ed.) [1979], List of problems, Homological group theory (C. T. C. Wall, ed.), *London Math. Soc. Lecture Notes* **36**, Cambridge University Press, Cambridge, 1979, 369–394.
- A. Weil [1952], Sur les théorèmes de de Rham, *Comment. Math. Helv.* **26** (1952), 119–145.
- J. E. West [1977], Mapping Hilbert cube manifolds to ANR’s: A solution of a conjecture of Borsuk, *Ann. of Math.* **106** (1977), 1–18.
- J. H. C. Whitehead [1949], Combinatorial homotopy, I, II, *Bull. Amer. Math. Soc.* **55** (1949), 213–245, 453–496.
- J. A. Wolf [1974], *Spaces of constant curvature* (3rd edition), Publish or Perish, Boston, 1974.
- H. J. Zassenhaus [1958], *The theory of groups*, 2nd ed., New York, Chelsea, 1958.

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